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MILITARY SPECIFICATION

POSITIONER, STEP

(For Torpedo Mk 37 Mod 2)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification describes the requirements for the procurement of a step positioner, NAVORD Dwg 1521127. The step positioner, as an element of a steering controller, serves to introduce electrical command signals, received from a source external to the steering controller, into a horizontal (azimuth) steering system of a torpedo in accurate increments of two degrees.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-S-642	Sodium Phosphate, Tribasic, Technical
FF-N-836	Nut, Hexagon and Square
FF-S-86	Screw, Cap, Socket-Head
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-S-365	Silver Plating (Electrodeposited)
QQ-S-571	Solder, Soft (Tin, Tin-Lead, and Lead-Silver)

FSC 1355

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QQ-Z-325	Zinc Plating (Electrodeposited)
TT-C-490	Cleaning and Pretreatment of Ferrous Surfaces and Organic Protective Coatings
TT-P-664	Primer Coating, Synthetic, Rust-Inhibiting Lacquer-Resisting
TT-T-291	Thinner; Paint, Volatile Mineral Spirits
PPP-B-621	Boxes, Wood, Nailed and Lock Corner

Military

MIL-P-116	Preservation, Methods of
MIL-B-857	Bolts, Nuts, Studs, and Tap Rivets (and materials)
MIL-S-901	Shockproof Equipment, Class HI (High Impact)
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification for
MIL-B-7883	Brazing of Steel, Copper, Copper Alloys and Nickel Alloys
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-8585	Primer Coating; Zinc-Chromate, Low Moisture Sensitivity
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111 for Metals)
MIL-P-15328	Coating, Pretreatment (Formula No. 117 for Metals)

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MIL-T-18404	Torpedoes, Environmental Requirements; General Specification for
MIL-E-82590	Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

DRAWINGS

Naval Ordnance Systems Command

Drawing 1521127	Step Positioner
Drawing 2190629	Steering Controller Assembly

PUBLICATIONS

Naval Ordnance Systems Command

OD 3000	Lubrication of Ordnance Material
OS 6341	Ordnance Design Requirements; General Miscellaneous

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

National Bureau of Standards Publications

Handbook H28	Screw-Thread Standards for Federal Services

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The step positioner, NAVORD Dwg 1521127, is an element of the steering controller in a horizontal steering system under gyro control. It angularly positions a shaft through 360 degrees of rotation in either direction in accurate increments of 2 degrees. The step positioner is so mounted on a bracket that its shaft is in alignment with and mechanically coupled to the shaft of a synchro-transformer, as shown on NAVORD Dwg 2190629. The step positioner is activated by electrical command signals, the direction of angular displacement being dependent upon the terminal pair energized (A and B or B and C).

3.2 General requirements

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the step positioner shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing step positioners that comply with the technical requirements of the contract. See 4.3.1.

3.2.2 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.2.3 Definitions. Definitions applicable to the environmental requirements of this specification are contained in MIL-T-18404.

3.2.4 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the step positioner shall conform strictly to the applicable Federal or Military specifications. Materials which are not covered by specification or which are not specifically designated herein shall be of the best quality, of the lightest practicable weight consistent with strength and durability, and entirely suited to the purpose for which intended under all conditions of test and service operation. Recommendations concerning the use of a superior substitute for any part or material specified herein shall be accompanied by a description of the proposed substitute, the reasons therefore, together with evidence to substantiate any claim as to its suitability. At the discretion of the Naval Ordnance Systems Command, test samples may be required to prove the suitability of the proposed substitute.

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3.2.4.1 Critical materials. Materials shall be selected on the basis of suitability and availability, taking into consideration the additional restrictions created during a time of national emergency. Critical materials shall be used only when essential for satisfactory operation of the Naval Ordnance Systems Command.

3.2.4.2 Magnesium. The use of magnesium requires the specific approval of the Naval Ordnance Systems Command or its appointed agency.

3.2.4.3 Flammable materials. Materials which will support combustion or which are capable of causing an explosion shall not be used.

3.2.4.4 Toxic materials. Materials which may produce harmful toxic effects under conditions encountered in the Naval Service shall not be used.

3.2.5 Moisture, fungus, and corrosion. The step positioner shall be moisture, fungus, and corrosion resistant in accordance with OS 6341.

3.2.6 Finishes and coatings

3.2.6.1 General. All machining operations shall be completed before the application of the corrosion-resistant treatment.

3.2.6.2 Plating. Electrodeposited metallic coatings shall conform to the following. The plating thickness shall in all cases be adequate to ensure conformance with the parts or equipment specification for conductivity and corrosion resistance.

Zinc	QQ-Z-325, Type II, Class 2 or 3
Nickel	QQ-N-290
Chromium	QQ-C-320, Type I
Silver	QQ-S-365

NOTE

Cadmium plating shall not be used.

3.2.6.3 Tin and lead-tin alloys. Platings of tin and lead-tin alloys, when used, shall be of adequate thickness and so continuous as to withstand exposure to simulated sea atmosphere as specified in MIL-T-18404.

3.2.6.4 Corrosion resistant steel

3.2.6.4.1 Cleaning. Remove all oil, grease, dirt, and other foreign matter by vapor degreasing, washing with Petroleum Spirits, TT-T-291, Grade 1, or by immersion in a boiling solution of Trisodium Phosphate,

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O-S-642, followed by rinsing in cold water and a final rinse in hot water. If necessary because of scale, immerse in the following solution, held at 60 degrees centigrade (C) (140 degrees Fahrenheit (F)), until scale is removed:

Nitric acid	20% by volume
Hydrofluoric acid	1 to 3% by volume
Water	77 to 79% by volume

Then rinse in cold water and blow dry with compressed air. If structures are fabricated with overlapping joints the preferred method of cleaning is by vapor degreasing to avoid entrapment of caustic or acid solutions.

3.2.6.4.2 Passivation. Immerse the clean metal in a solution of equal parts of nitric acid (40 degrees Baume) and water held at a temperature of 71 degrees C (160 degrees F) for five minutes. Rinse first in clean cold water followed by a rinse in hot water. Then blow dry with compressed air.

3.2.6.5 Anodic film. Parts fabricated from aluminum or aluminum alloys shall be anodized in accordance with MIL-S-8625 or MIL-C-5541.

3.2.6.6 Magnesium and magnesium alloys. Magnesium and magnesium alloys, when authorized, shall be finished in accordance with MIL-M-3171.

3.2.6.7 Painting. If painting is selected as the final finish, the following processes shall be used:

3.2.6.7.1 Cleaning. Surfaces to be painted shall be freed of dirt, dust, grease, mill scale, welding flux, and other foreign matter. Cleaning may be accomplished with volatile grease solvent, vapor degreasing detergents, hot alkali cleaning solutions, or other processes which have no deleterious effect on the surfaces. Welding irregularities shall be removed. Alkali cleaning shall not be used on aluminum or aluminum alloy surfaces. When alkali cleaning is used, the final rinse shall consist of a dilute chromic acid solution (0.5 pound of chromic acid per 100 gallons of water). When fabrication involves overlapping joints, vapor degreasing is the preferred method of cleaning.

3.2.6.7.2 Pretreatment. Either of the following processes may be used:

(a) One coat of wash pretreatment in accordance with MIL-P-15328, applied as a continuous film 0.0004 to 0.0005 inch thick, or

(b) A hot-dip tank phosphate or chromate-phosphate treatment conforming to TT-C-490, Grade I.

3.2.6.7.3 Primer. One coat of primer in accordance with TT-P-664 or MIL-P-8585 (for aluminum or magnesium surfaces), applied as a continuous film 0.0004 to 0.001 inch thick.

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3.2.6.7.4 Enamel. Two coats of enamel, light gray, MIL-E-15090, Class 2, applied as a continuous film approximately 0.001 inch thick.

3.2.6.7.5 Exceptions. The following surfaces shall not be painted: mounting surfaces and shaft extensions.

3.2.7 Marking. Marking shall be in accordance with NAVORD Dwg 1521127. Marking shall be by means of engraving, metal stamping, embossing, or by permanently affixed nameplate.

3.2.7.1 Marking data. Unless otherwise specified in the contract or order, or detailed in the applicable drawings, marking shall include the following:

(a) In area B, NAVORD Dwg 1521127: name of item or assembly; serial number; NAVORD drawing number and drawing revision letter; operating voltage and current; manufacturer's name, symbol, or trademark; manufacturer's part or assembly number (if desired).

NOTE

The date and place of manufacture shall not be shown.

(b) In area A, NAVORD Dwg 1521127: terminal designations.

(c) Index marks as shown on NAVORD Dwg 1521127.

3.2.7.2 Serial numbers. Serial numbers of each major unit or part shall start with number one and continue consecutively for the total quantity to be supplied regardless of the number of contracts or orders involved. Confirmation of the use of serial numbers under any contract or order shall be obtained from the Naval Ordnance Systems Command or its appointed agency. Where the same major unit or part may have been previously manufactured by another contractor, information concerning serial numbers shall be obtained from the Naval Ordnance Systems Command or its appointed agency.

3.2.8 Threaded parts. Bolts, nuts, screws, studs, etc., shall conform to the requirements of FF-N-836, FF-S-86, or MIL-B-857. Screw threads for all threaded parts shall be in accordance with Screw-Thread Standards for Federal Services, Handbook H28, and shall be of size 2-56, 4-40, 6-32, 8-32, 10-32, 1/4-20, or 1/4-28. If larger sizes are needed, they shall be of the coarse-thread series. Threaded parts (other than bolts, screws, studs, nuts, etc.) requiring threads of special diameter and pitch shall employ the special pitch of 16 threads to the inch. For applications where a finer pitch is necessary, one of the following shall be selected: 20, 28, 36, 44, or 56.

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3.2.8.1 Thread projection. Screws or bolts secured by nuts or other retaining devices shall be of such length that a minimum of 1-1/2 threads and a maximum of 1/8 inch plus 1-1/2 threads for screws up to one inch in length, and 1/4 inch plus 1/1-2 threads for screws over one inch in length, shall project beyond the retainer. These requirements shall not be construed to preclude the use of screws assembled in blind holes, spacers, etc., or where design restrictions require that the threaded portion of the screw be flush with the retaining device. In no case shall the screw be clipped, bent, or otherwise shortened after plating.

3.2.8.2 Thread engagement. Screws and bolts in tapped parts other than in nuts shall enter and engage for a distance at least equal to the diameter of the screw or bolt.

3.2.8.3 Self-tapping screws. Self-tapping screws will not be acceptable except where employed in approved components and where their removal from the component is a remote possibility.

3.2.8.4 Locking devices. Locking devices shall be provided for all screw thread assemblies except for captive screws on withdrawable units. The working loose during tests or in service of threaded parts will be considered a defect in design. All lockwashers shall conform to the applicable Military Standards and shall be fabricated of corrosion resistant steel, plated phosphorbronze, or steel zinc-plated, in accordance with QQ-Z-325, Type II, Class 2 or 3.

3.2.8.4.1 External tooth type lockwashers shall not be used with screws 1/4 inch or greater in diameter.

3.2.8.4.2 Spring type lockwashers may be used for screws of all sizes.

3.2.9 Lubrication. Provision shall be made to adequately lubricate all bearings and wearing surfaces, with due consideration to the operation of the step positioner in tropical and arctic climates. The design shall be such that satisfactory operation will result, with a minimum of re-lubrication, and that a minimum of grades and types of lubricants are required. OD 3000 shall be used when selecting lubricants and related materials. Previous to production of the step positioner the contractor may be required to submit to the Naval Ordnance Systems Command for approval a list of lubricants selected for use and the application for which each lubricant is intended.

3.2.10 Methods of wiring

3.2.10.1 No more than three wires, including wires from parts, shall terminate at any one terminal.

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3.2.10.2 Pressure contacts. Electrical connections shall not depend upon wires, lugs, terminals, and the like clamped between a metallic and an insulating or pliable material. Connections of this type shall be clamped between rigid metallic members, such as in an assembly of a machine screw, two nuts, and two washers. If such an assembly cannot be used and maintenance of a tight connection depends upon the resistance of an insulating or pliable material to compressive stress or shear, the connection shall be securely soldered.

3.3 Construction and design

3.3.1 General

3.3.1.1 Objective. The basic design objectives are that the equipment will meet the needs of the Naval Service and that the final product will reflect the utmost in simplicity, have maximum reliability consistent with the state of the art, be easy to operate and maintain, and be economical to produce.

3.3.1.2 Needs of the Naval Service. The equipment shall be designed to withstand storage and service conditions over long periods without benefit of overhaul and with limited maintenance.

3.3.1.3 Reliability. It is imperative that reliability of operation be considered of prime importance.

3.3.1.4 Simplicity. The equipment shall represent the simplest design consistent with the operational and environmental conditions set forth in this specification.

3.3.1.5 Operation and maintenance. The equipment shall be designed so that it can be easily operated and maintained. Accessibility and serviceability features which will lead to simplified maintenance shall be a prime consideration of accessory and auxiliary equipment.

3.3.2 Dimensions and tolerances. Dimensions and tolerances of the step positioner shall conform with those shown on NAVORD Dwg 1521127.

3.3.3 Weight. The weight of the step positioner shall not exceed one pound.

3.3.4 Mechanical design

3.3.4.1 The shaft shall be free to rotate through 360 degrees in either direction.

3.3.4.2 Shaft wobble shall not exceed 0.001 inch total indicator reading.

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3.3.4.3 Dimensional relationship between the shaft and the mounting holes of the step positioner shall conform with those shown on NAVORD Dwg 1521127.

3.3.5 Electrical design

3.3.5.1 Three wire-hook terminals, A, B, and C, shall be provided and shall be mounted in a glass insulator located on the end opposite to the shaft extension. None of these terminals shall be electrically connected to the frame of the step positioner. Terminals shall be hot tin dipped or solder tinned.

3.4 Performance and product characteristics. The step positioner shall meet the following performance and product characteristics:

3.4.1 Mechanical

3.4.1.1 The angular positional error at any integral point of rotation shall be not greater than 0.20 degree when referred to any other integral point of rotation.

3.4.1.2 A detent shall prevent rotation in either direction of rotation due to an externally applied torque of up to 12 ounce-inches.

3.4.2 Electrical

3.4.2.1 The insulation breakdown voltage of the step positioner shall be not less than 500 volts rms at 60 Hertz (Hz) when applied for a period of one minute between the frame and any terminal.

3.4.2.2 The step positioner shall operate over a voltage range of 20 to 30 volts dc, with the duration of the applied voltage not longer than 1/12 second.

3.4.2.3 The step positioner, with an applied voltage of 30 volts dc, shall operate on a current which shall not exceed one ampere at a temperature of plus 77 degrees F and shall not exceed 1.4 amperes at a temperature of minus 20 degrees F.

3.4.2.4 A nominal 25 volts dc, independent of polarity, when applied across terminals A and B shall cause the shaft of the step positioner to rotate in a clockwise direction when looking at the shaft extension end of the step positioner. When this voltage is applied across terminals B and C the shaft shall rotate in a counterclockwise direction.

3.4.2.5 The step positioner shall be capable of operating at a minimum rate of 6 steps per second with an inertial load of 0.68 plus or minus 0.04 ounce-inch added to the shaft of the step positioner.

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3.4.2.6 The step positioner shall develop a minimum torque of not less than 5 ounce-inches squared during each step.

3.4.2.7 Mechanical hangup. After being supplied by an aiding torque of 20 ounce-inches minimum (as described in 4.5.4.4.4), the step positioner shall perform as specified in 4.5.4.4.3.

3.4.2.8 Insulation resistance. The insulation resistance of the step positioner shall be not less than 10 megohms between any terminal and the enclosing case under any service or environmental condition.

3.4.3 Operating life. The step positioner shall meet the performance and other requirements during and after operation for not less than 250,000 steps in each direction of rotation under the environmental conditions described in 4.5.2.

3.5 Environmental requirements. The step positioner shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.4.

3.5.1 Vibration. The step positioner shall meet the following vibration requirements:

3.5.1.1 Vibration, nonoperating. The step positioner shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.1.

3.5.1.2 Vibration, operating. The step positioner shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.4.1.2.

3.5.1.3 Vibration, transportation. The step positioner shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.3.

3.5.2 Shock, transportation. The step positioner shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.

3.5.3 Temperature and humidity

3.5.3.1 High temperature and humidity, nonoperating. The step positioner shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.3.1.

3.5.3.2 High temperature, operating. The step positioner shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.3.2.

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3.5.3.3 Low temperature, nonoperating. The step positioner shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.3.3.

3.5.3.4 Low temperature, operating. The step positioner shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.3.4.

3.6 Workmanship. The step positioner, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

3.6.1 Welding. All welds shall be of ample size with thorough penetration and good fusion, and shall be substantially free of injurious scabs, blisters, abnormal pock-marks, bruises or other surface defects. Welds shall be capable of withstanding a strength test to destruction in which failure shall not occur within the weld itself. There shall be no evidence of harmful effects due to the welding process on the material adjacent to the weld. All welds shall be cleaned of scale, oxidation products, and excess flux by sand blasting, wire brushing, buffing, or other suitable means. Surfaces cleaned with acid shall be completely neutralized. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength. At least two welds shall be used to hold any part.

3.6.2 Soldering. Only class SN60 solder in accordance with QQ-S-571 shall be used for electrical connections. Soldering processes shall conform to the requirements specified in MIL-S-6872. No assembly shall depend solely on soft solder for mechanical strength except for relatively light parts which have by actual use proved to be suitable.

3.6.3 Brazing. All brazed joints shall be in accordance with MIL-B-7883, and shall be of ample size to meet the individual requirements of the parts concerned.

3.6.4 Riveting. Riveting operations shall be carefully performed in order to ensure that the rivets are tight and satisfactorily headed.

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3.6.5 Cleanup. Prior to and after final assembly all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, rosin flash, and other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the step positioner shall be classified as follows:

(a) Preproduction inspection (paragraph 4.4)

(b) Quality conformance inspection (paragraph 4.5)

(c) Periodic production inspection (paragraph 4.4)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five step positioners, manufactured in accordance with 3.2.1, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the step positioner by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.5, a random sample of five step positioners will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specifi-

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cation, and drawings. The step positioners shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.4.1 Vibration

4.4.1.1 Vibration, nonoperating. The positioner shall be subjected to to the "Vibration, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.5.1.1.

4.4.1.2 Vibration, operating. The step positioner shall be subjected to the "Vibration, operating" test of MIL-E-82590(OS) and during and thereafter shall meet the requirements of 3.5.1.2.

4.4.1.3 Vibration, transportation. The step positioner shall be subjected to the "Vibration, transportation" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.5.1.3

4.4.2 Shock, transportation. The step positioner shall be subjected to the "Shock, transportation" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.5.2.

4.4.3 Temperature and humidity

4.4.3.1 High temperature and humidity, nonoperating. The step positioner shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.5.3.1.

4.4.3.2 High temperature, operating. The step positioner shall be subjected to the "High temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.5.3.2.

4.4.3.3 Low temperature, nonoperating. The step positioner shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.5.3.3.

4.4.3.4 Low temperature, operating. The step positioner shall be subjected to the "Low temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.5.3.4.

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4.4.4 Operating life. Using the interrupter or square wave generator (4.5.3.7), operate the step positioner for not less than 250,000 steps in each direction of rotation, varying the voltage between 20 and 30 volts, and repeatedly changing the direction of rotation. The stepping rate shall be approximately 6 steps per second. The environmental conditions during the test shall be as described in 4.5.2.

4.4.5 Waiver of tests. Tests specified in 4.4.1.3 and 4.4.2 above may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.4.6 Functional tests. At the conclusion of the tests specified in 4.4.1.3 and 4.4.2 above, functional and other tests, as applicable, shall be conducted on the step positioner to determine freedom from operational malfunction and compliance with all performance requirements and product characteristics of this specification.

4.5 Quality conformance inspection. Each step positioner delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirements specified herein shall be considered cause for rejection.

4.5.1 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit the performance of the required quality conformance tests.

4.5.2 Test conditions. Unless otherwise specified, the step positioner shall be subjected to quality conformance tests under the following conditions:

4.5.2.1 Temperature. Room ambient 65 degrees F to 95 degrees F (18 degrees C to 35 degrees C).

4.5.2.2 Altitude. Normal ground.

4.5.2.3 Vibration. None.

4.5.2.4 Humidity. Room ambient to 95 percent relative maximum.

4.5.3 Items of test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

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4.5.3.1 An instrument capable of measuring the angular position of a small shaft to within an accuracy of 0.20 degree.

4.5.3.2 A torque watch gage capable of measuring up to 20 ounce-inches with an accuracy of plus or minus one percent.

4.5.3.3 A source of 500 volt, rms, 60 Hz, ac power.

4.5.3.4 A source of dc power adjustable between 20 and 30 volts.

4.5.3.5 A dc ammeter, scale 0 to 5 amperes, accurate to plus or minus one percent.

4.5.3.6 A lever switch, nonlocking, double-pole single throw each side, center-off, rated at 3 amperes and 150 watts.

4.5.3.7 An interrupter or square wave generator and sensitive relay capable of making and breaking the step positioner circuit at a rate of 6 steps per second.

4.5.3.8 A megohmmeter, rated at 500 volts dc, with a midscale reading of not less than 10 megohms.

4.5.4 Test procedure

4.5.4.1 Angular error. Mount the step positioner in the test instrument (4.5.3.1). Following the instructions of the manufacturer of the test instrument, determine the angular error of the step positioner in each of the 180 step positions of the shaft. Measure the errors for rotation of the shaft in both the clockwise and counterclockwise direction of rotation. The step positioner shall meet the requirements specified in 3.4.1.1.

4.5.4.2 Torque

4.5.4.2.1 Static torque. Mount the gage (4.5.3.2) on the shaft of the step positioner. Adjust the gage to indicate a torque of 21 ounce-inches. The step positioner shall not step when an external torque of up to 12 ounce-inches is applied to the shaft. Repeat this test for nine additional and widely separated angular positions of the shaft of the step positioner.

4.5.4.2.2 Dynamic torque. Mount the torque gage on the shaft of the step positioner and apply a retarding torque of at least 5 ounce-inches squared to the direction of rotation. With a single electrical pulse applied, the step positioner shall advance 2 degrees to meet the requirements of 3.4.2.6. Repeat the foregoing test for nine additional and widely separated angular positions of the shaft of the step positioner.

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4.5.4.3 Insulation. Apply the ac voltage (4.5.3.3) across the frame of the step positioner and each of terminals A, B, and C. The step positioner shall withstand the application of this voltage without damage, arcing, or breakdown.

4.5.4.4 Operation. Connect one terminal of the dc power source, (4.5.3.4), through the ammeter (4.5.3.5), to terminal B of the step positioner. Connect the other terminal of the power source through the lever switch (4.5.3.6) to each of terminals A and C.

4.5.4.4.1 Voltage. Test the operation of the step positioner in each direction of rotation throughout the operating voltage range of 20 to 30 volts. Stepping shall be positive and accurate.

4.5.4.4.2 Current. During the test of 4.5.4.4.1, observe the reading of the ammeter. Current shall not exceed 1 ampere at plus 77 degrees F.

4.5.4.4.3 Stepping. Connect the step positioner to an inertial load of 0.68 plus or minus 0.04 ounce-inch with the lever switch (4.5.3.6) or the interrupter (4.5.3.7), and operate the step positioner at the rate of 6 steps per second in such a manner as to cause stepping in both directions. The step positioner shall follow the applied electrical impulses positively and accurately.

4.5.4.4.4 Mechanical hangup. Mount the gage (4.5.3.2) on the shaft of the step positioner. With a single electrical pulse, applied and held, adjust the gage to indicate an aiding torque of 20 ounce-inches minimum in the direction of rotation of the shaft. Release the electrical pulse, maintaining the aiding torque of 20 ounce-inches minimum. After removing the electrical pulse, remove the aiding torque. The step positioner shall then perform as specified in 4.5.4.4.3. Tests to be conducted in both directions of rotation.

4.5.4.5 Insulation resistance. With the megohmmeter (4.5.3.8), measure the insulation resistance between each terminal of the step positioner and the enclosing case. The step positioner shall meet the requirements specified in 3.4.2.8.

4.5.4.6 Visual and dimensional examinations. Each step positioner shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.5.4.7 Packaging, packing, and marking. Examine the packaging, packing, and marking to determine conformance with the requirements of Section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The step positioner shall be packaged in accordance with Method IID of MIL-P-116.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The step positioner shall be packaged in accordance with Method 1C-1 of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The step positioner, packaged as described in 5.1.1.1, shall be packed in a wooden box in accordance with PPP-B-621, Class 2.

5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 95.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The step positioner, packaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Levels A or C cushioning. The unit package shall incorporate sufficient cushioning material, bracing or other adequate shock absorbing devices to ensure that the step positioner will meet all the requirements of Section 3 after being tested in accordance with 4.4.1.3 (Vibration) and 4.4.2 (Shock). At the time of packaging or packing, the moisture content of the cushioning material shall be not more than 5 percent.

5.4 Marking

5.4.1 Special markings. None, unless otherwise specified.

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5.4.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. The step positioner is intended for use in Torpedo Mk 37 Mod 2.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.2.1 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID*
4.4	Preproduction and periodic production inspection	-

DIDS (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDS when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

Custodian:
Navy - OS

Preparing Activity
Navy - OS
Project No. 1355-N193

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-P-19277B(OS) Positioner, Step.

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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Standardization Division (QA3)
Indian Head, MD 20640

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